VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

22BAD0136

Final Assessment Test(FAT) - Apr/May 2025

Programme	B.Tech.	Semester	Winter Semester 2024-25
Course Code	BCSE427L	Faculty Name	Prof. Arunkumar
Course Title	Cognitive Robotics	Slot	D1
		Class Nbr	CH2024250502394
Time	3 hours	Max. Marks	100

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: Understand how our psychology and neuroscience understanding of behavior and intelligence informs the design of robotics models and applications

CO2: Compare, select and apply different machine learning methods for intelligent behavior in robots

CO3: Analyze the methods and software/hardware technologies for robotics research and applications

CO4: Discuss the state of the art in cognitive and intelligent robotics models, and how this leads to the design of future robot applications

Section - I

Answer all Questions (4 × 10 Marks)

01. A fully autonomous smart car is required to be designed for an educational institution to handle the need for mobility of students within the campus. By considering this scenario how different smart car mobility-based dynamics are handled, elucidate the various agent types and explain about its functions.

[10] (CO1/K2)

02. While carrying out randomized path planning for a mobile robot used in vacuum cleaning applications, the robot must be able to identify obstacles and take an alternate paths. Discuss elaborately how a visibility graph can be used to resolve the above-mentioned problem with appropriate diagram.

[10] (CO2/K2)

03. A mobile robot is used in construction site of a multistorey building for carrying the construction material. Suggest a suitable Simultaneous Localization and Mapping (SLAM) based method in step by step manner for implementing in the mobile robot to handle the unknown environment. Illustrate your answer with a neat diagram.

[10] (CO3/K3)

04. While using multiple mobile robots in the heavy mechanical component manufacturing industrial applications with the help of python robot programming method discuss in detail about the development of an obstacle avoidance in mobile robotic system.

[10] (CO4/K3)

Section - II

Answer all Questions (4 × 15 Marks)

05. Highlight the relevance of embodied cognition in a garbage collecting robot and elaborate different aspects of its modelling (8 marks).
Give a detailed account of the level of abstraction with neat diagram for above mentioned scenario (7 marks).

[15] (CO1/K2)

06. Consider a mobile robot used for conducting search and rescue operations across multiple terrain surfaces in a region affected by a natural disaster. While capturing image feeds in such an environment, appropriate visual perception and recognition methods are required.
a) Identify and explain the optimized methods suitable for visual perception and recognition in this context. (6 marks)
b) Describe in detail the image capturing process for both humans and machines. Discuss the associated visual perception models. Support your explanation with suitable diagrams. (9 marks)

[15] (CO2/K2)

07. Construct a 2 -Dimensional map building algorithm in step-by-step manner with appropriate pseudo code for the given diagram as shown below in Figure-1.

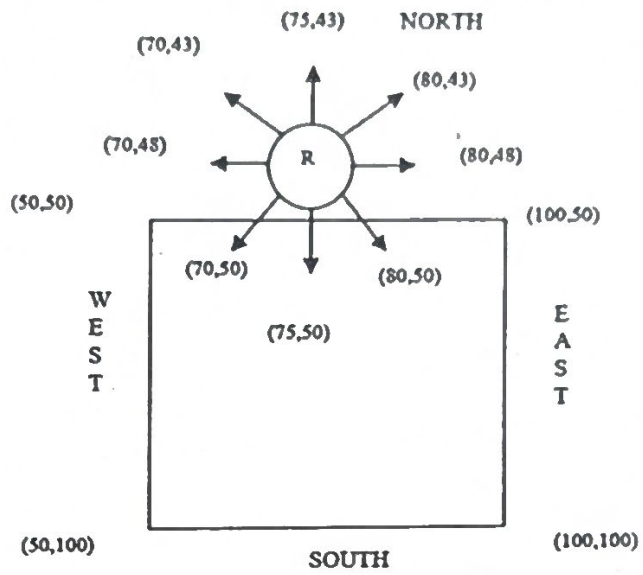

Figure-1

[15] (CO3/K3)

08. In textile industries for moving the raw material inside the large production plant mobile robots are used nowadays, appropriate selection of mobile robot components are most important for its efficient operation. List out the basic type of sensors, controllers, actuators and communication modules used in those type of mobile robots (6 Marks) Elaborate the working principle of your proposed system with a neat diagram. (9 Marks)

[15] (CO4/K2)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)